Developing an ANN-Powered R Shiny Application for Real-Time Seizure Prediction from EEG Data

Kim, M.A.[1] & Marsellos, A.E.[2]

[1]Manhasset High School, 200 Memorial Place, Manhasset, NY, USA

[2]Dept. Geology, Environment and Sustainability, Hofstra University, Hempstead, NY, USA

Abstract

Electroencephalography (EEG) is a non-invasive technique used to measure the brain's electrical activity through sensors placed on the scalp. Widely used in both clinical and research settings, EEG provides real-time insights into brain function and is especially valuable in diagnosing certain neurological functions, particularly epilepsy. Seizures produce distinctive changes in EEG signals, making EEG a valuable tool for identifying and predicting seizure activity.

This research aims to develop a seizure predicting model using EEG data and machine learning techniques implemented in R. EEG files were meticulously selected from PhysioNet, a research resource for public EEG signals. Once the signals were downloaded, they were band-pass through 0.5 - 40 Hz and re-referenced in order to removed noise extract the relevant features. Various signal processing tools are applied to isolate patterns that typically characterize the seizures. After completing the extraction and plotting EEG signal amplitude (voltage) in real time, machine learning was then applied to predict certain patterns. Random Forest and Artificial Neural Networks (ANN) were the two programs used to train the model and predict certain patterns and peaks that characterize the seizures. With further enhancement, the model was trained to distinguish the difference between a seizure and non-seizure state.

The ability to predict seizures in real time has profound implications for patient safety and quality of life. By alerting patients and caregivers before a seizure occurs, several accidents can be avoided. This research also paves way for other neurological conditions, which can be identified using EEG signals and further be processed using machine learning to minimize accidents.

Background Information

- Electroencephalography (EEG) records the brain's electrical activity using sensors place on the scalp
- EEG measures voltage fluctuations caused by ionic currents in neurons
- It is non-invasive, low cost, and provides high resolution
- EEG is used to diagnose conditions like epilepsy, sleep disorders, seizures, and brain injuries
- EEG is increasingly used to investigate chemo brain by detecting changes in brainwave patterns
- Alterations in specific EEG frequency bands (alpha/theta) may reflect cognitive impairment post-chemotherapy
- Studying EEG in this context may lead to biomarkers for early detection and monitoring of chemo brain

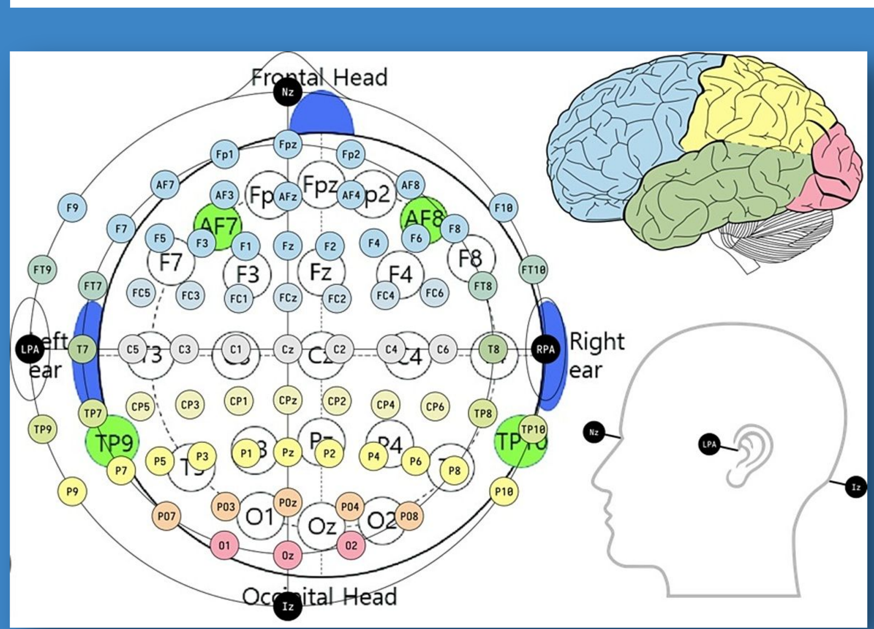

Figure 1: Visual map of the 10-20 system

Methods

- **Acquire Data**
  - Download selected EEG files and seizure annotations
- **Pre-Process**
  - Band-pass 0.5-40 Hz, notch 60 Hz, re-reference, segment into 10 s windows
- **Feature Extraction**
  - Time (mean, variance) & frequency (band power, spectral analysis) per window
- **Model Training**
  - Using Artificial Neural Networks and Random Forest
- **Visualization**
  - Importance plot, correlations or coherence between brain regions
- **App Importation**
  - Use of Shiny to import code as an app

Results

```
model_ann <- neuralnet(y ~ x, data = nn_df, hidden = 1, linear.output = TRUE)
```

- A simple feedforward artificial neural network (ANN) was trained to predict the next time step of the EEG signal using a single hidden layer
- The input (x) was a scaled EEG signal; the output (y) was the signal shifted by one time step to model temporal dependencies
- After training 999 data points, the ANN demonstrated its ability to track the EEG trend in short-term predictions
- Visualization showed close alignment between actual and predicted values, confirming the ANN's ability to learn temporal patterns

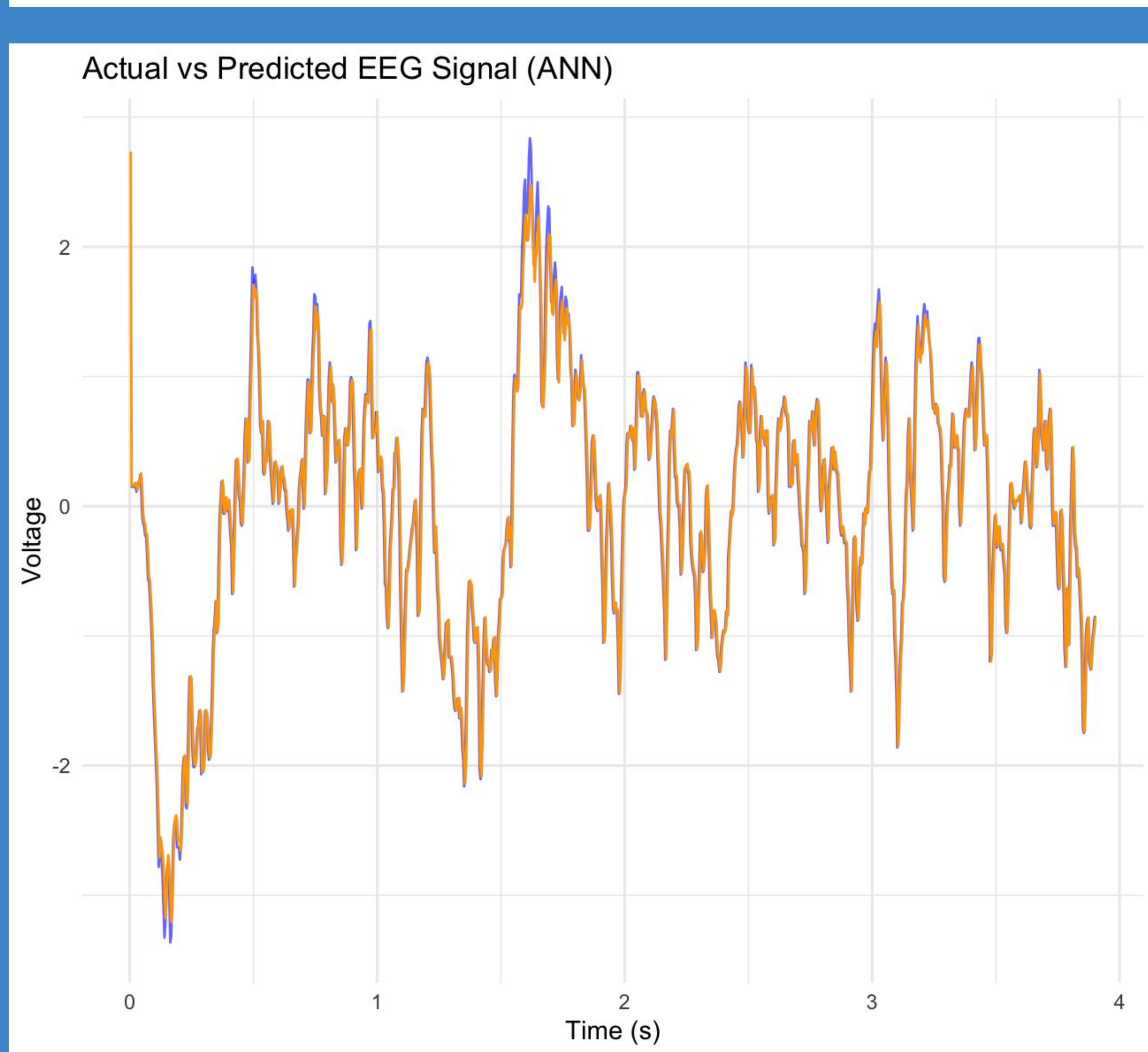

Figure 2: Visual plot of machine learning: ANN used to predict respective EEG signal

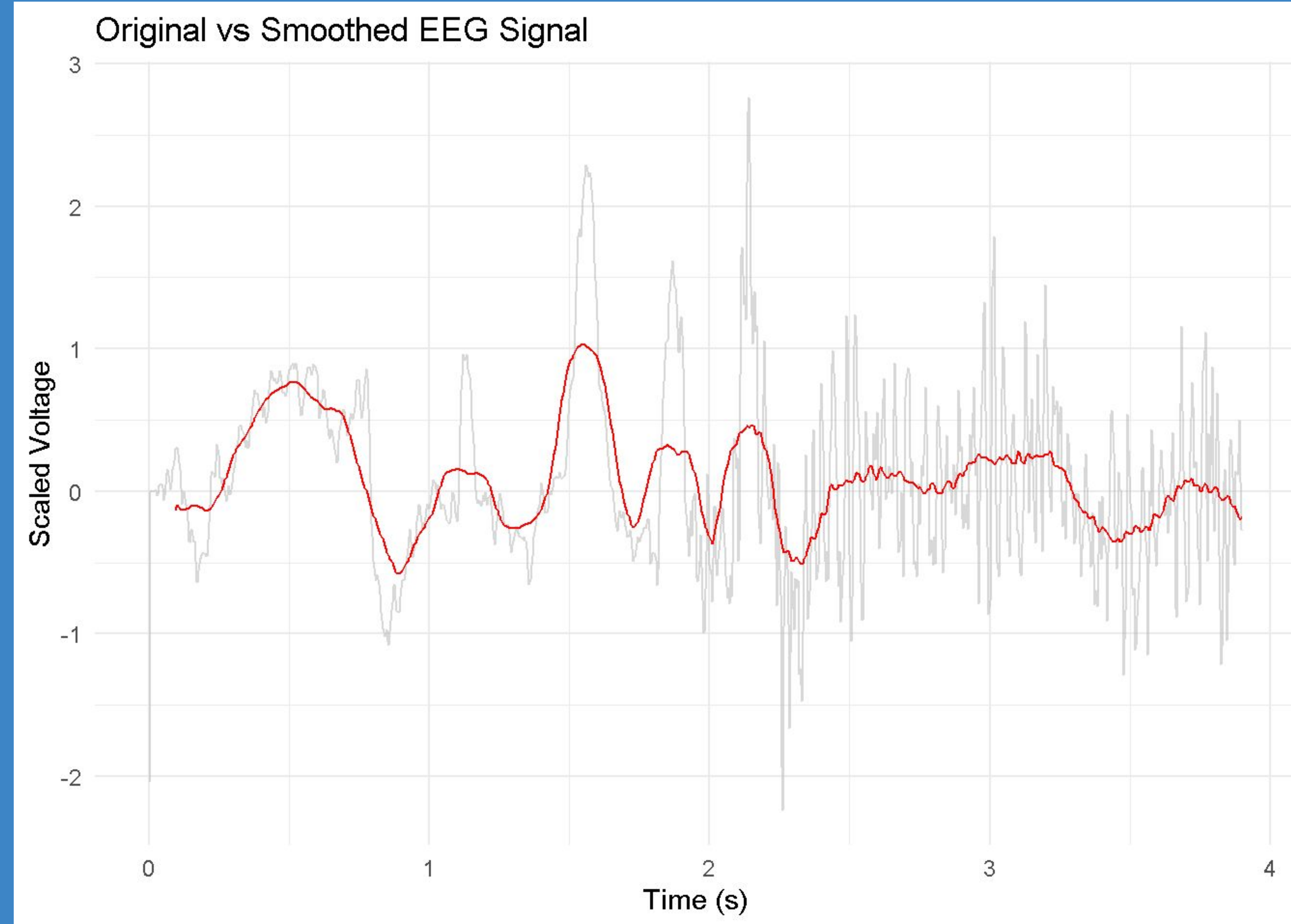

Figure 3: Plotting of the original vs smoothed EEG signal (filtered out of the noise)

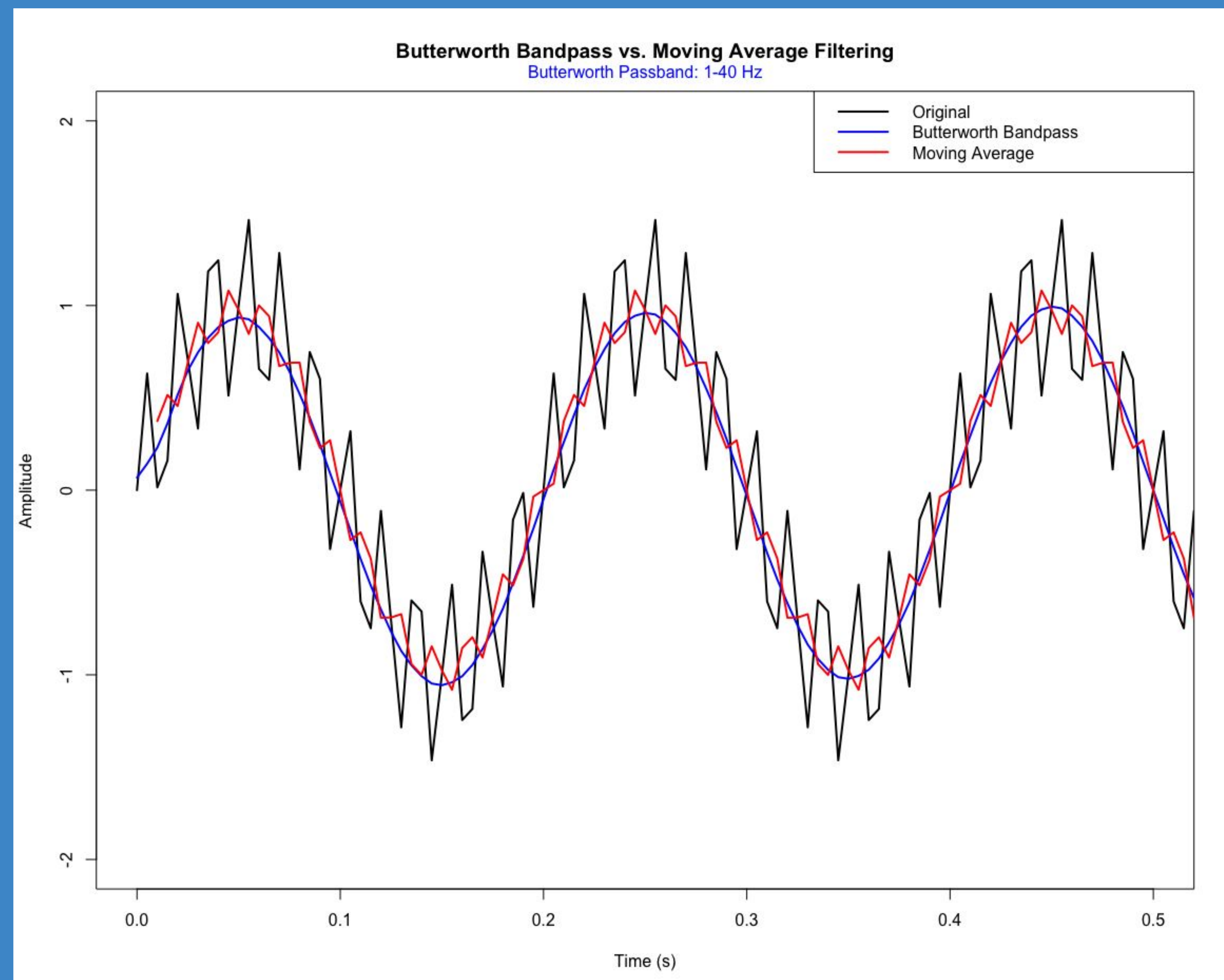

Figure 4: Use of the butterworth bandpass filter (only pass 1-40 Hz), removes 60 Hz component of Gamma Waves and higher

Discussion/Conclusion:

- The Artificial Neural Network (ANN) successfully learned dependency in EEG signals, showing close alignment between predicted and actual brain activity
- The model demonstrates short-term seizure prediction potential, which is critical for early warning systems about patient safety
- Band-pass filtering (0.5-40 Hz) and 60 Hz noise removal were effective in isolating relevant EEG features for machine learning analysis
- Feature extraction in both time and frequency domains improved the model's ability to detect partners characteristic of seizure onset
- The ANN's performance suggests that real-time seizure prediction apps using Shiny in R are a feasible future direction
- This approach can be extended to other neurological disorders, marking early detection beyond epilepsy

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References:

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